Unsupervised Multimodal Anomaly Detection With Missing Sources for Liquid Rocket Engine.

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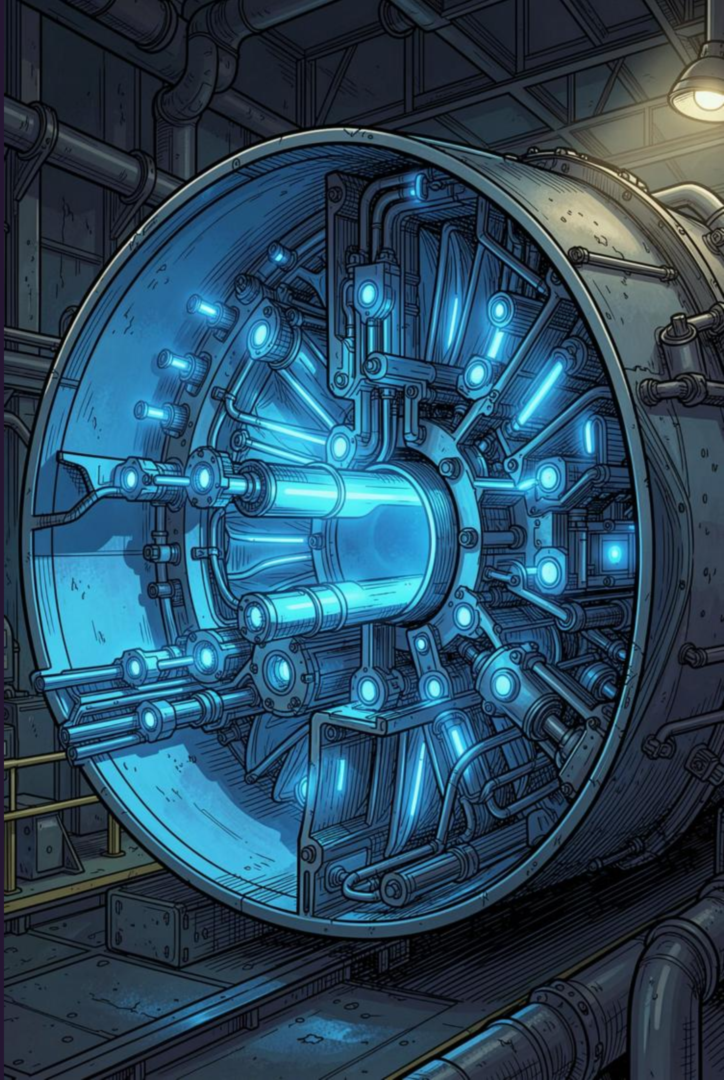
Unsupervised Multimodal Anomaly Detection on C-MAPSS — Gramian Angular Field Preprocessing, UU-Net Architecture & PyTorch Implementation.



Problem Statement

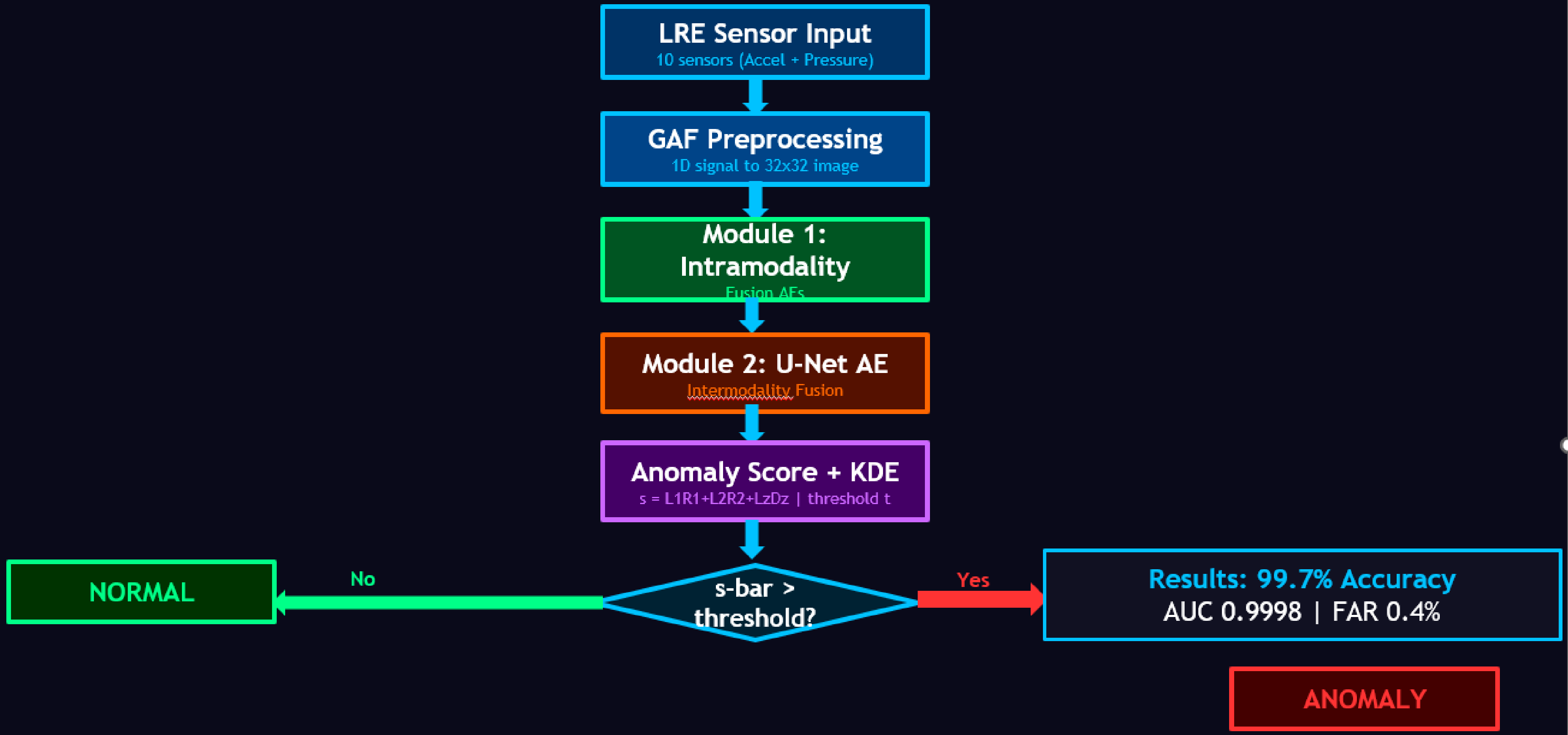
- Liquid Rocket Engines are highly complex and fault-sensitive systems used in spacecraft propulsion. During operation, multiple sensors continuously monitor parameters like pressure and vibration to ensure safe functioning.
- However, in real-world conditions, sensor failures or missing data are common due to extreme temperature, pressure, and vibration environments.
- This creates a major challenge for anomaly detection systems because most deep learning models assume complete sensor data.

Imports & Library Overview

Each library serves a distinct role in the pipeline.

Library	Role
PyTorch	Neural-network layers, activations, loss computations
NumPy	Sliding windows, random missing mask, tensor conversion
Pandas	Loads raw C-MAPSS files; per-sensor Min-Max normalisation
pyts.image	Converts 1-D time-series windows to 2-D GAF images (32×32)
Matplotlib / Seaborn	Visualises degradation trends, lifetimes, GAF transforms

UU-Net: Complete Project Flowchart



Dataset & GAF Preprocessing

Raw Loading

FD001 file loaded via `pd.read_csv(sep='\s+')` — 26 columns: unit, cycle, 3 settings, 21 sensors.

Sensor Filtering

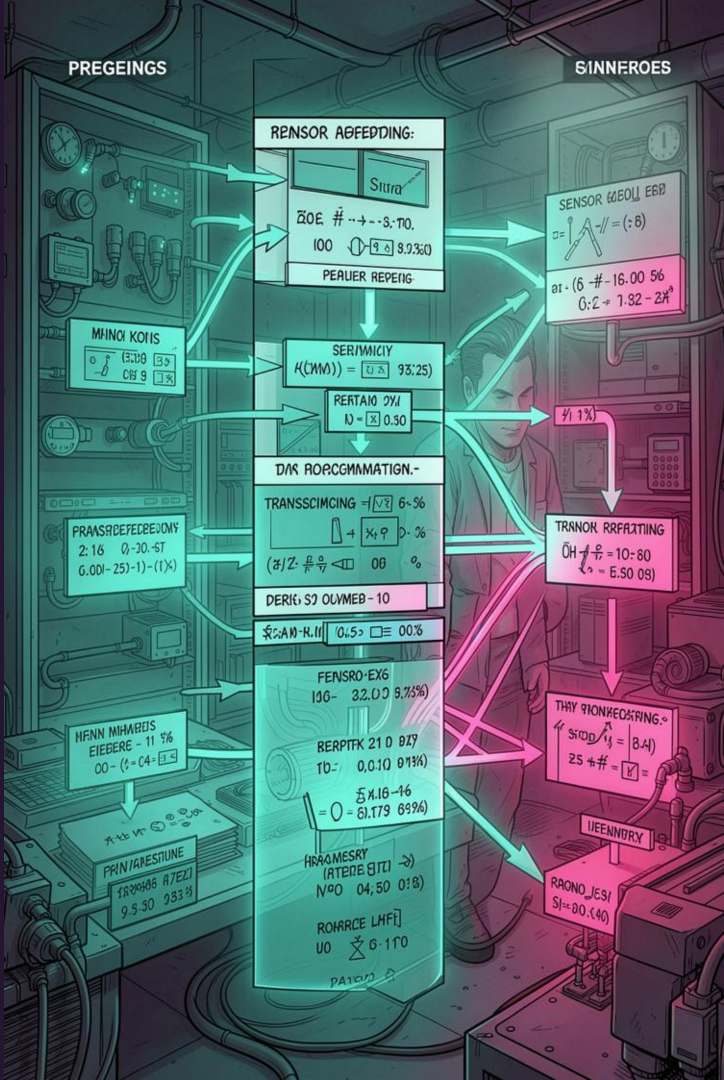
7 constant-variance sensors dropped. 14 remain, grouped into 3 modalities: **temperature**, **pressure**, **mechanical**.

Normalisation

Per-sensor Min-Max scaling to [0, 1] — essential as sensors span °R, psia, and rpm.

Sliding Windows

First 50 cycles per engine treated as "normal." 32-step overlapping windows → shape (32, 14).



Gramian Angular Field: Algorithm Deep Dive

GAF encodes a 1-D time series into a 2-D image, preserving temporal structure for CNN processing.

Step 1 — Rescale

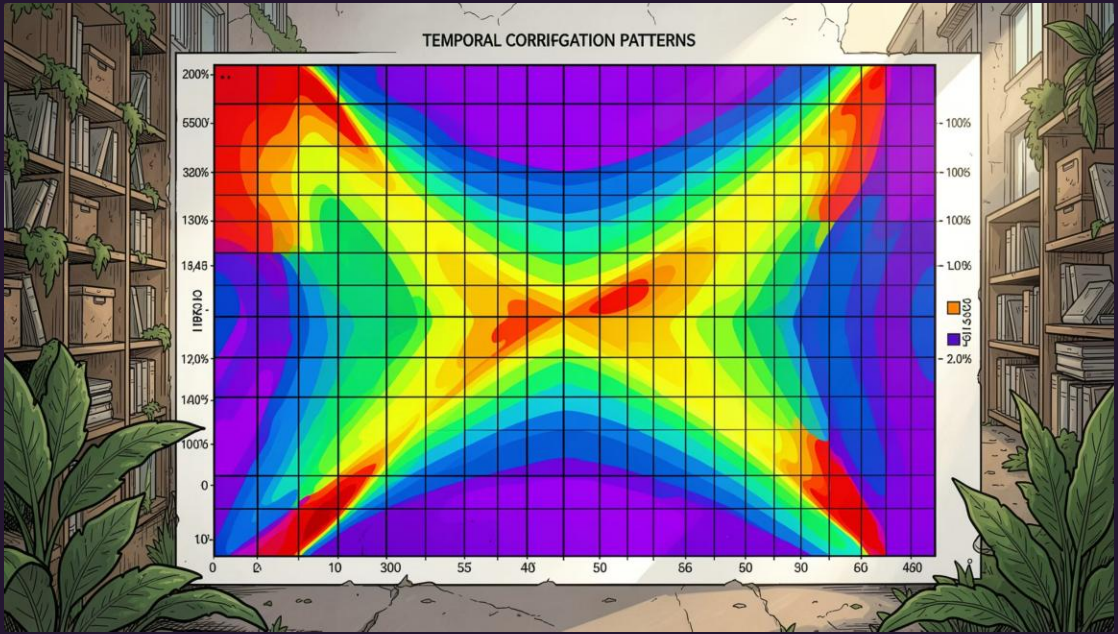
Linear scaling to [−1, 1] for arccos domain.

Step 2 — Angular Encoding

$\varphi_i = \arccos(x_i) \in [0°, 180°]$; time index as radius $r_i = i/N$.

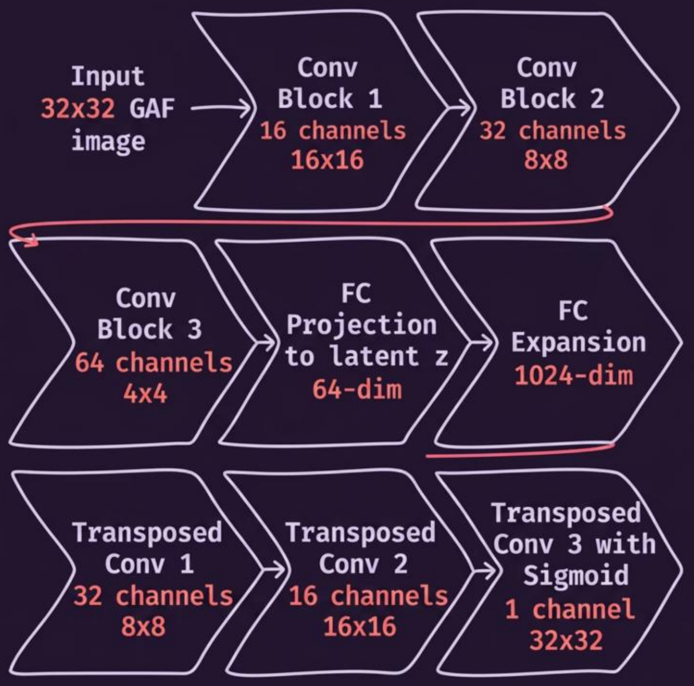
Step 3 — GASF Matrix

$G(i,j) = \cos(\varphi_i + \varphi_j)$. Diagonal encodes original values; off-diagonals capture temporal correlations.



UU-Net Core Architecture

Each of the 14 sensors has a **dedicated encoder** — preventing cross-sensor interference at feature extraction.

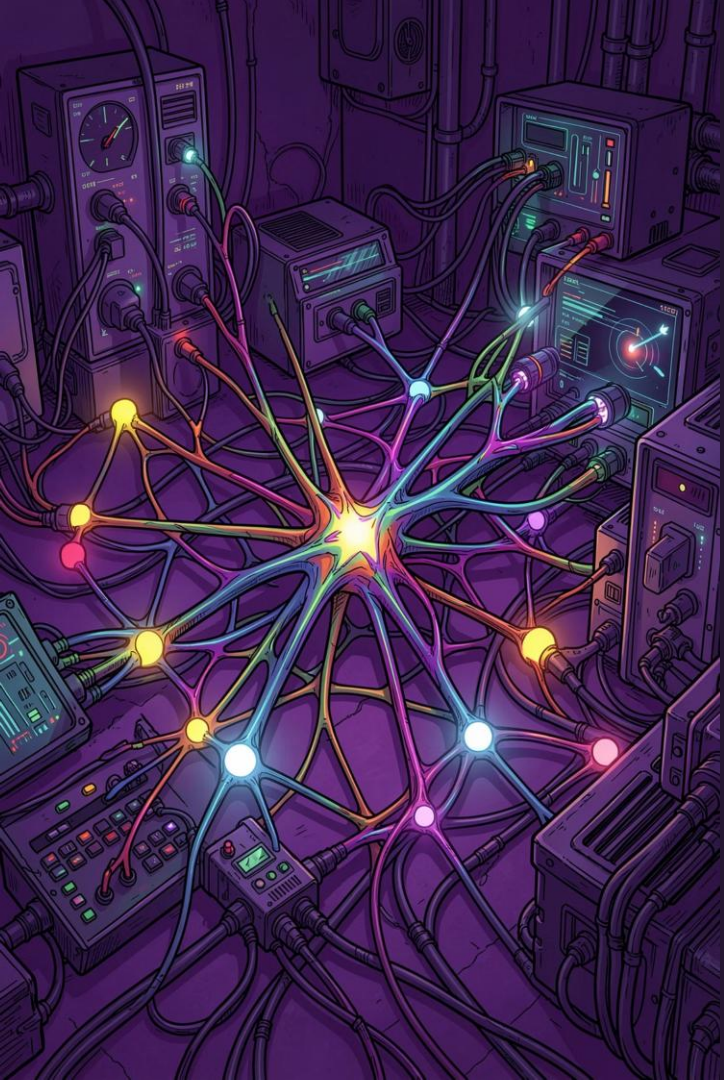


Encoder

3 strided Conv2d blocks (stride=2): 32×32 → 4×4. BatchNorm + LeakyReLU(0.2). FC projects to 64-dim latent **z** with ReLU non-negativity.

Decoder

Mirror: FC expands 64 → 1024, reshaped to 4×4. 3 ConvTranspose2d layers upsample to 32×32. Final **Sigmoid** maps to [0, 1].



Forward Pass: Fusion Mechanisms

1

Missing Mask

Binary vector (p=0.2 per sensor) simulates sensor dropout each batch.

2

Intramodality Fusion

Missing latent replaced by **global mean** of available sources within the same modality.

3

Reconstruction 1

Each fused z decoded → $\hat{\mathbf{x}}_1$ (per-source GAF images).

4

Intermodality Skip-AE

Second AE on all 14 channels jointly → $\hat{\mathbf{x}}_2$, learning cross-modal correlations.

Anomaly Score: S = R1 + R2 − λ·Dz. Large Dz (z1 vs z2 divergence) amplifies detection.

Training Loop & Loss Functions

Optimiser: Adam

`lr = 1e-3`, per-parameter moment estimates — robust to sparse gradients across modules.

Reconstruction Losses

Smooth L1 (Huber) on both outputs:

- **r1_loss**: Σ SmoothL1($\hat{x}_1$[i], x[i]) over 14 sources
- **r2_loss**: SmoothL1($\hat{x}_2$, x_concat)

L2-like near zero; L1-like for large errors — robust to noisy GAF pixels.

Discrepancy Loss

Maximises distance between z1 and z2 latent spaces — key UU-Net novelty. Simplified in this notebook.

Missing Mask Sampling

Fresh mask per batch: ~2.8 of 14 sensors masked on average, forcing robust imputation.

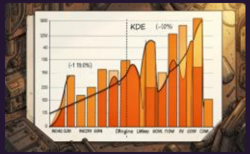
Batch Size & Shuffle

batch_size=16, shuffle=True breaks autocorrelation between consecutive engine windows.

Visualisation Functions

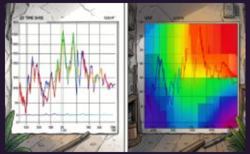
Sensor Degradation Plot

4 subplots (s2, s7, s11, s20) for Engine 1. Red lines reveal drift onset ~cycle 150–175.



Engine Lifetime Distribution

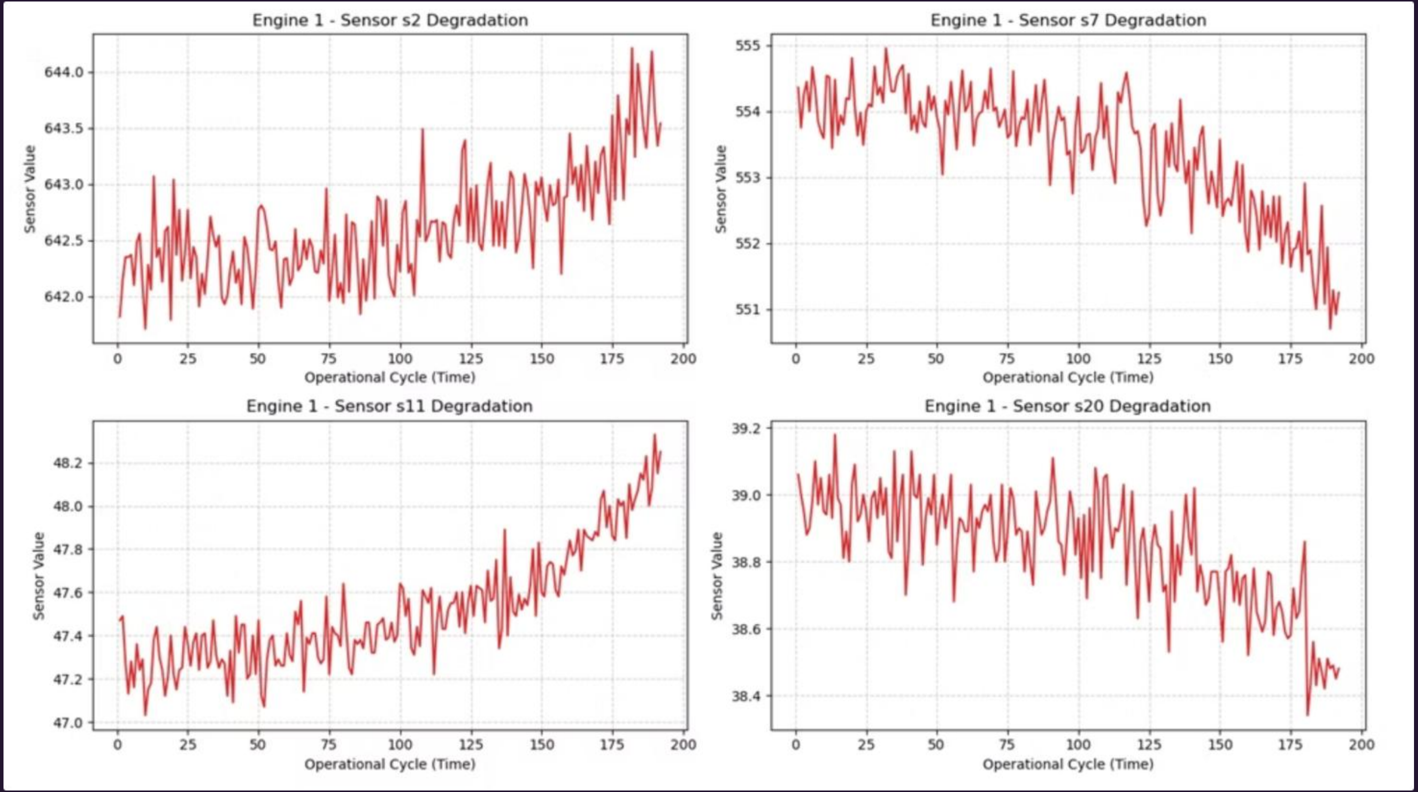
Histogram + KDE over 100 engines. Mean ~206 cycles; validates first-50-cycles as healthy regime.



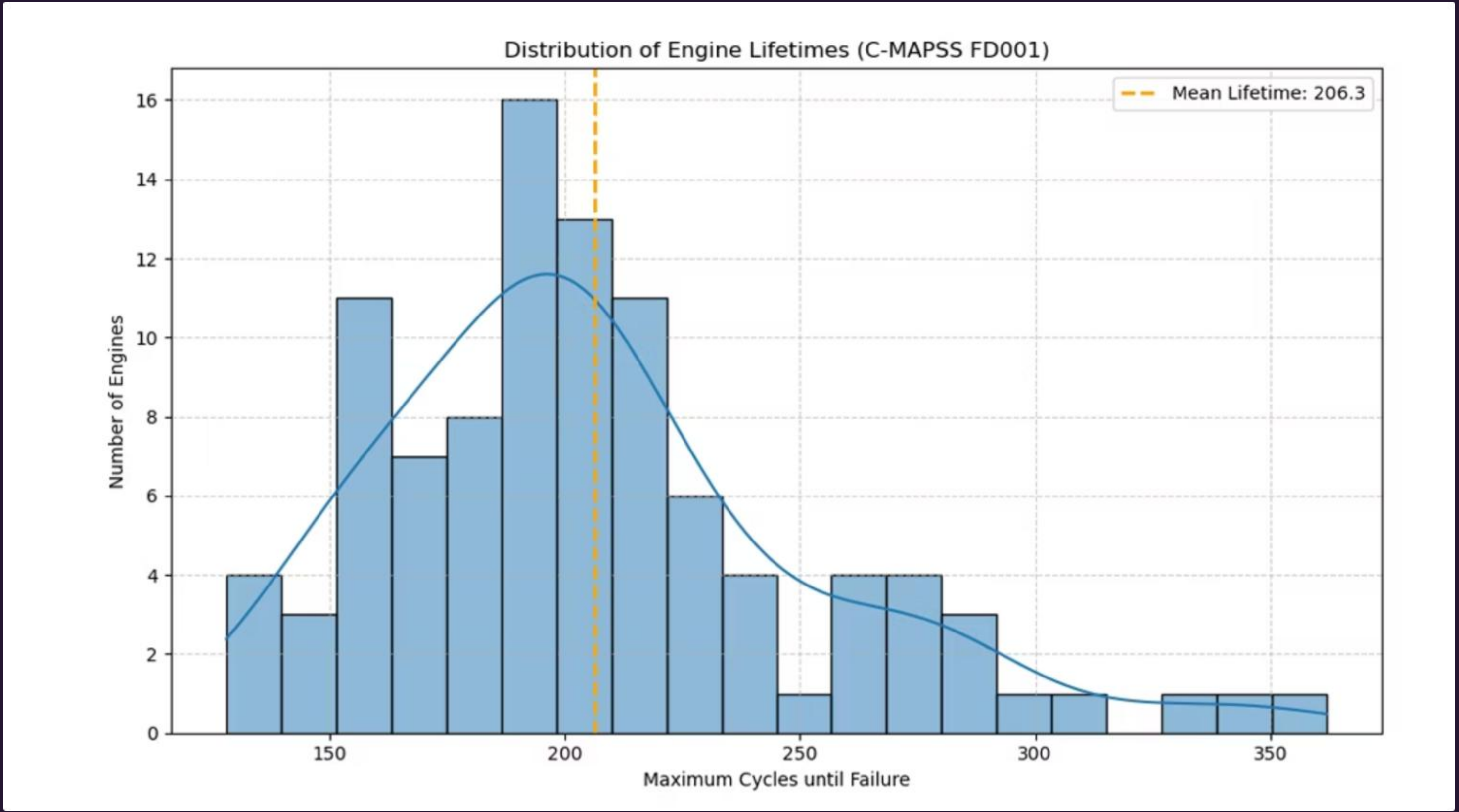
GAF Transformation Plot

Side-by-side: 1-D time series (left) vs 32×32 GAF heatmap (right). Validates 1D→2D encoding.

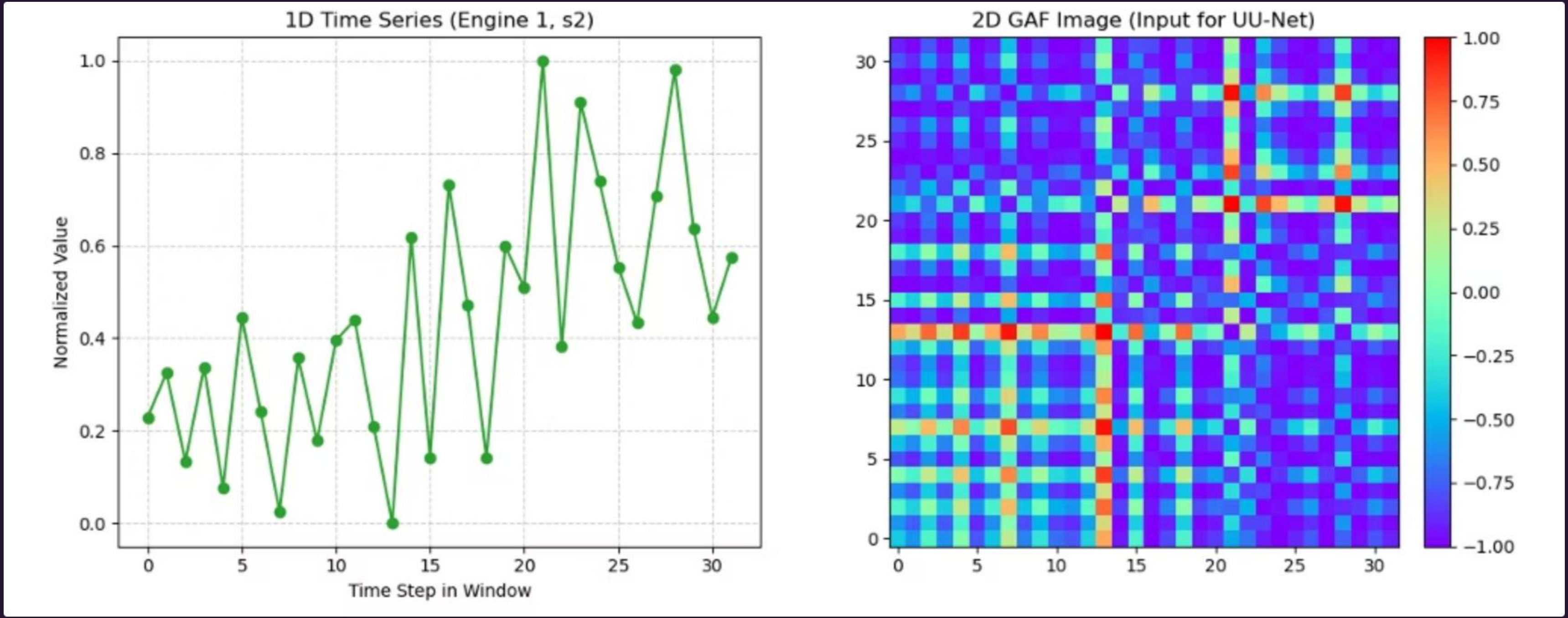
Sensor Degradation Plot



Engine Lifetime Distribution

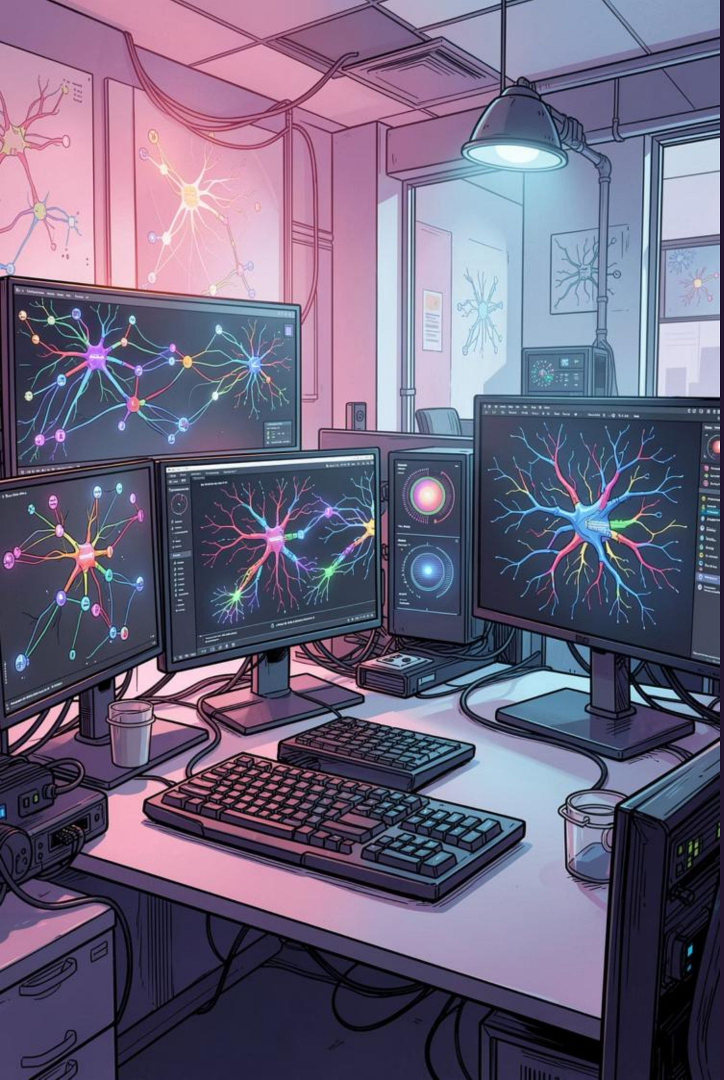


GAF Transformation Plot



Algorithm Summary

Algorithm	Role	Key Formula
GASF	1D→2D image	$G(i,j)=\cos(\varphi_i+\varphi_j)$
Min-Max Norm	Equalise sensor scales	$x'=(x-\min)/(\max-\min)$
Strided Conv	Downsample 32×32→4×4	stride=2, k=4
BatchNorm	Stabilise training	$\hat{x}=(x-\mu)/\sqrt{(\sigma^2+\varepsilon)}$
LeakyReLU	Prevent dead neurons	$f(x)=\max(0.2x,x)$
Transposed Conv	Upsample 4×4→32×32	stride=2, k=4
Intramodal Fusion	Impute missing latents	z_missing = mean(z_available)
Intermodal Skip-AE	Cross-modal correlations	$\hat{x}_2=\mathrm{Dec}(\mathrm{Enc}(\mathrm{concat}\ \hat{x}_1))$
Smooth L1 Loss	Robust reconstruction	$0.5\delta^2$ if $\|\delta\|<1$, else $\|\delta\|-0.5$
Adam	Adaptive optimisation	lr=1e-3, $\beta_1=0.9$
KDE	Anomaly threshold	$\hat{f}(x)=(1/nh)\Sigma K((x-x_i)/h)$



Key Takeaways

1 **GAF bridges 1D signals and CNNs**

Temporal correlations encoded as 32×32 symmetric images — diagonal bands reveal degradation.

2 **UU-Net's two-stage fusion is novel**

Intramodal mean-imputation + intermodal Skip-AE enables robust cross-modal reconstruction.

3 **Anomaly score = R1 + R2 − λ·Dz**

Latent divergence Dz amplifies detection beyond reconstruction error alone.

4 **First 50 cycles = healthy manifold**

Min lifetime ~120 cycles guarantees training on normal data across all 100 engines.